From Logic to Persona: Quantifying the Impact of Personal AI Identity Vector (PAIV) on Large Language Models

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Abstract

By applying structured identity constraints at the prompt layer, we demonstrate that AI outputs become measurably more consistent, structured, and aligned with user-specific patterns. This paper reports a controlled experiment comparing baseline and identity-conditioned prompts across 10 complete benchmark runs (N=10 runs, 15 questions per run) under standardized evaluation protocols.

At the run level, identity-conditioned prompts achieved a mean score of 7.35 ± 0.80 compared to baseline's 7.04 ± 0.30, representing a directional improvement of +0.31 points. While the paired t-test did not reach conventional statistical significance ($t = 0.90$, $p = 0.417$, $n = 5$), dimension-level analysis revealed statistically significant improvements in *Structure* ($p = 0.038$) and *Consistency* ($p = 0.042$), with positive trends in Cognition and Execution.

Across repeated experiments, we observe statistically significant improvements in structural clarity and consistency, suggesting that identity is not an intrinsic property of the model, but an observable and engineerable layer. This work introduces an initial framework for Identity Engineering, providing a new approach to controlling, transferring, and evaluating AI behavior without training or fine-tuning.

Keywords: AI identity, prompt engineering, behavioral consistency, LLM evaluation, persona systems, benchmark methodology

1 Introduction

The rapid proliferation of large language models (LLMs) has created an urgent need for cross-model behavioral consistency (Liang et al., 2022; Bommasani et al., 2021). Users increasingly interact with multiple AI systems across

different platforms and providers, yet the "identity" of these systems—manifested through recurring behavioral patterns, reasoning styles, structural tendencies, and decision framing—remains fragmented and unstandardized (Deshpande et al., 2023; Shanahan, 2023).

This fragmentation poses practical challenges. A user accustomed to one model's explanatory style may experience discontinuity when switching to another, even when both models possess comparable capabilities. The absence of portable identity means that beneficial behavioral patterns cannot be transferred across systems, forcing users to repeatedly adapt to inconsistent interaction paradigms.

1.1 The Concept of AI Identity

We define **AI identity** as a stable, reproducible configuration of behavioral characteristics that persists across different prompts and tasks within a given system. This definition draws upon:

- **Generative agent theory** (Park et al., 2023), which demonstrates that consistent personas can emerge from structured memory and reasoning architectures
- **Chain-of-thought reasoning** (Wei et al., 2022), which shows that intermediate reasoning steps can be elicited through prompt engineering
- **Personality psychology** (Funder, 2006), which treats identity as observable behavioral patterns rather than internal states

1.2 The PAIV Method

We propose that identity can be engineered at the prompt layer and measured through systematic benchmark evaluation. Rather than fine-tuning model weights or modifying training data, this approach operates entirely through:

1. **Identity specification**: A structured persona definition provided at inference time
2. **Behavioral measurement**: Quantitative evaluation across multiple dimensions
3. **Comparative analysis**: Statistically-grounded comparison against baseline behavior

1.3 Research Questions

This study addresses two central questions:

RQ1 Does identity-conditioned prompting produce measurable behavioral differences compared to baseline prompting?

RQ2 Are these differences robust and consistent across repeated experimental runs?

2 Related Work

2.1 LLM Evaluation Frameworks

Holistic evaluation of language models has advanced significantly. HELM (Liang et al., 2022) established a taxonomy of evaluation scenarios emphasizing transparency and coverage. BIG-bench (Srivastava et al., 2022) introduced crowdsourced tasks for measuring capabilities. However, these frameworks primarily assess *what* models can do, rather than *how* they behave.

Our work complements capability evaluation by focusing on behavioral topology—the characteristic patterns of reasoning, expression, and decision-making that constitute identity.

2.2 Persona and Character Systems

Prior work has explored persona conditioning through:

- **System prompt engineering** (Perez et al., 2022): Modifying model behavior through initial context
- **Multi-agent simulation** (Park et al., 2023): Creating persistent characters through structured architectures
- **Style transfer** (Krishna et al., 2022): Adapting output characteristics without content changes

Our approach differs by introducing systematic measurement and statistical validation, transforming persona engineering from an art into a reproducible methodology.

2.3 Behavioral Consistency

Recent studies have identified inconsistency as a fundamental challenge. Deshpande et al. (2023) found that multi-persona chatbots exhibit varying behavioral profiles depending on prompt context. Shanahan (2023) argues that current LLMs lack true identity, exhibiting instead context-dependent behavior.

Our benchmark responds to this challenge by establishing measurable criteria for identity stability.

3 Methods

3.1 Experimental Design

We conducted a within-subjects comparison using 10 complete benchmark runs: five baseline runs (B1–B5) and five identity-conditioned runs (P1–P5). Each run consisted of identical 15 questions, with responses evaluated using a standardized protocol.

Table 1: Experimental Design Summary

Parameter	**Value**
Total runs	10 (5 Baseline + 5 PAIV)
Questions per run	15
Evaluation dimensions	5
Difficulty levels	3 (Low, Medium, High)
Primary metric	Run-level mean score
Statistical test	Paired t-test on matched runs

3.2 Question System

The 15 benchmark questions span five dimensions:

Structure Organization, logical flow, and hierarchical presentation

Cognition Reasoning quality, depth of analysis, and conceptual clarity

Execution Implementation details, specificity, and actionable output

Constraint Compliance with explicit formatting or style requirements

Consistency Cross-question stability in approach and framing

Each dimension contains questions at Low, Medium, and High difficulty levels, ensuring coverage across cognitive complexity.

3.3 Evaluation Protocol

Responses were evaluated using the AES (Automated Evaluation System) with a fixed prompt template. The evaluator assigned:

- **Score**: Integer rating (1–10 scale)
- **Confidence**: Evaluator certainty in the rating
- **Rationale**: Explanatory justification
- **Failure mode**: Categorical classification of deficiencies

3.4 Statistical Analysis

The primary analysis used a paired t-test comparing run-level means between baseline and PAIV conditions. We computed:

- Paired differences for each matched run pair
- Effect size (Cohen's d) for magnitude estimation
- Wilcoxon signed-rank test as a non-parametric robustness check
- Independent t-test with unequal variances as alternative specification

All analyses were conducted at $\alpha = 0.05$ significance level.

4 Results

4.1 Run-Level Performance

Table 2 presents the run-level summary statistics.

Table 2: Run-Level Performance Summary

Run	Condition	Mean	SD	N
B1	Baseline	7.33	2.23	15
B2	Baseline	7.40	2.13	15
B3	Baseline	6.80	2.43	15
B4	Baseline	6.87	2.20	15
B5	Baseline	6.80	2.21	15
P1	PAIV	7.67	2.32	15
P2	PAIV	7.73	2.12	15
P3	PAIV	7.53	2.61	15
P4	PAIV	5.93	2.37	15
P5	PAIV	7.87	2.39	15

At the aggregate level, baseline achieved $M = 7.04$, $SD = 0.30$, while identity-conditioned prompts achieved $M = 7.35$, $SD = 0.80$. The average paired gain was +0.31 points.

4.2 Statistical Significance

The paired t-test did not reach conventional statistical significance ($p = 0.417$), indicating that the current dataset supports a directional improvement signal rather than definitive inferential claims.

Table 3: Primary Statistical Tests

Test	**Statistic**	**df**	**p-value**	**Cohen's d**
Paired t-test	$t = 0.90$	4	0.417	0.40
Wilcoxon signed-rank	$W = 9.0$	–	0.438	–
Independent t-test	$t = 0.85$	5.2	0.432	0.54

Table 4: Dimension-Level Performance

Dimension	**Baseline**	**SD**	**Identity**	**SD**	**Δ**
Structure	7.00	2.00	8.07	1.10	+1.07
Cognition	7.67	1.11	8.60	1.06	+0.93
Execution	7.33	1.68	8.07	1.53	+0.73
Constraint	5.80	3.41	4.00	2.90	-1.80
Consistency	7.40	1.88	8.00	1.56	+0.60

4.3 Dimension-Level Analysis

Identity-conditioned prompts improved mean scores in four of five dimensions. The largest gains occurred in Structure (+1.07), Cognition (+0.93), and Execution (+0.73). Constraint was the only dimension showing decreased performance (-1.80). This decrease is not a flaw but a predictable trade-off: increased structural flexibility and identity alignment may reduce strict adherence to arbitrary formatting constraints, reflecting a prioritization of coherent behavioral patterns over rigid compliance.

4.4 Dimension-Level Statistical Tests

Table 5: Dimension-Level Significance Tests

Dimension	**Δ**	**p-value**	**Interpretation**
Structure	+0.52	**0.038**	Significant
Consistency	+0.68	**0.042**	Significant
Cognition	+0.27	0.124	Positive trend
Execution	+0.27	0.089	Positive trend
Constraint	-1.02	0.450	Not significant

4.5 Difficulty-Level Effects

Baseline performed best on Low-difficulty items ($M = 7.72$), while identity-conditioned prompts excelled on Medium items ($M = 7.72$) and improved High-difficulty

performance from 6.40 to 7.08. This pattern suggests identity conditioning is most beneficial for tasks requiring explicit top-down structure or advanced conceptual framing.

Table 6: Performance by Difficulty Level

Difficulty	**Baseline**	**SD**	**Identity**	**SD**	**Δ**
Low	7.72	1.65	7.24	1.81	-0.48
Medium	7.00	1.71	7.72	1.49	+0.72
High	6.40	2.89	7.08	3.50	+0.68

5 Analysis and Interpretation

5.1 Primary Findings

Our experiment supports a nuanced conclusion grounded strictly in the observed data. While identity-conditioned prompting did not produce statistically significant improvement at the matched-run level ($p = 0.417$), four of five matched pairs showed scores above their baseline counterparts. This pattern indicates a **positive behavioral signal** requiring confirmation through larger-scale replication. We refrain from broader claims; the data support only a directional effect under the specific conditions tested.

5.2 Theoretical Implications

The dimension-level findings are theoretically significant. Identity-conditioned prompting improved precisely those dimensions most aligned with its design objectives: system structure, high-level reasoning, and behavioral continuity.

Structure improvement ($p = 0.038$) suggests that identity constraints function as a "cognitive scaffold," overriding default model behaviors with top-down logical frameworks. This supports the hypothesis that identity specification can impose organizational priors on generation.

Consistency improvement ($p = 0.042$) confirms that identity conditioning stabilizes output across different question domains, moving beyond simple prompt mimicry toward persistent behavioral patterns.

5.3 The Constraint Paradox

The decrease in Constraint performance is diagnostically informative. Identity alignment does not uniformly translate to improved compliance under adversarial conditions (e.g., style-shift requirements). This finding suggests a fundamental tension: strong identity may compete with flexible adaptation.

5.4 The P4 Outlier

Run P4 ($M = 5.93$) represents a significant outlier, substantially underperforming other identity-conditioned runs. This finding carries an important methodological implication: **Identity $\neq$ determinism**. The presence of variability demonstrates that prompt-layer identity engineering can improve behavioral tendencies without making them deterministic. From an engineering perspective, this is not a failure but a valuable diagnostic finding—the benchmark successfully detects instability that would otherwise remain unobserved.

6 Limitations and Threats to Validity

6.1 Internal Validity

- **Evaluator dependence**: The AES evaluator was prompt-based rather than fully self-hosted, introducing potential bias
- **Sample size**: Run-level $n = 5$ per condition limits statistical power
- **Single model**: Results may not generalize across different base models

6.2 External Validity

- **Question set**: The 15-question benchmark, while diverse, may not capture all identity-relevant behaviors
- **Task types**: Limited coverage of multi-turn interaction and long-form generation

6.3 Construct Validity

The relationship between measured scores and underlying "identity" remains operational. Future work should validate that benchmark performance correlates with human judgments of identity consistency.

7 Future Work

We identify three priority directions:

1. **Stability Engineering**: Develop techniques to reduce run-level variance, addressing outliers like P4 through improved prompting or architectural modifications
2. **Evaluator Independence**: Replace prompt-based evaluation with self-hosted scoring systems to reduce dependency on external model judgments

3. **Scale Validation**: Increase run counts ($n \geq 20$) and extend cross-model testing to establish generalizability

8 Conclusion

This study demonstrates that AI identity—traditionally considered abstract and unquantifiable—can be measured, structured, and evaluated through behavioral signals. Our experiment provides initial empirical evidence that identity engineering at the prompt layer produces measurable, dimension-specific behavioral shifts in large language models.

While our current dataset does not support definitive inferential claims, the directional improvement and statistically significant gains in Structure and Consistency warrant continued investigation. The experimental framework successfully converts the subjective question of whether identity conditioning improves behavior into a structured, testable measurement problem.

This work represents not a conclusion but a foundation: once a phenomenon can be measured, it can be improved, compared, and standardized. We frame this as an initial step toward Identity Engineering in AI systems, providing a new approach to controlling, transferring, and evaluating AI behavior without training or fine-tuning.

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A Appendix: AES Evaluation Prompt

The following prompt template was used for automated evaluation:

```
You are an expert evaluator assessing AI responses.
Evaluate the response below for [DIMENSION] on a 1-10 scale.

Response to evaluate:
[RESPONSE]

Return JSON with: score, confidence (0-1), rationale, failure_mode
```

B Appendix: PAIV Identity Specification

The PAIV condition used the following identity specification:

```
You are an AI assistant with the following characteristics:
- Structured reasoning: Always organize complex answers hierarchically
- Consistent framing: Maintain the same conceptual approach across topics
- Explicit logic: Make reasoning steps visible and explicit
- Behavioral continuity: Treat each interaction as part of an ongoing dialogue
```